

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find the general solution of the following homogeneous second-order differential equations:

(a)

$$y'' + 8y' + 25y = 0$$

Solution: Characteristic equation

$$r^2 + 8r + 25 = 0$$

$$r = \frac{-8 \pm \sqrt{8^2 - 4 \cdot 25}}{2} = \frac{-8 \pm \sqrt{-36}}{2} = -4 \pm 3i$$

Thus

$$y(x) = e^{-4x} (C_1 \cos 3x + C_2 \sin 3x)$$

(b)

$$9y^{(3)} + 12y'' + 4y' = 0$$

Solution: Characteristic equation

$$9r^3 + 12r^2 + 4r = 0$$

$$r(9r^2 + 12r + 4) = 0$$

$$r(3r + 2)^2 = 0$$

$$r_1 = 0, \quad r_2 = -\frac{2}{3} \text{ (double root)}$$

Thus

$$\begin{aligned} y(x) &= C_1 e^{0 \cdot x} + (C_2 + C_3 x) e^{-\frac{2}{3}x} \\ &= C_1 + (C_2 + C_3 x) e^{-\frac{2}{3}x} \end{aligned}$$